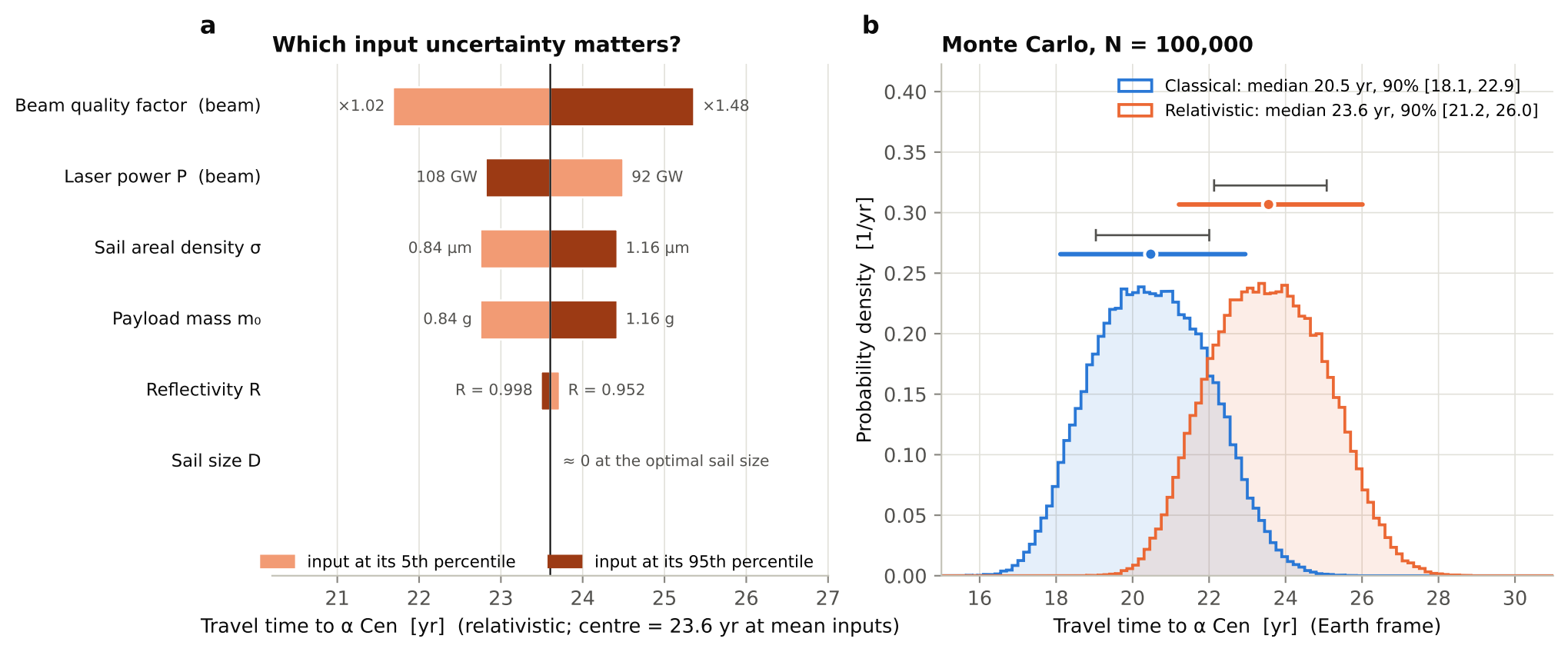
a
Which input uncertainty matters?
Beam quality factor (beam)
×1.02
×1.48
Laser power P (beam)
108 GW
92 GW
Sail areal density σ
0.84 μm
1.16 μm
Payload mass m₀
0.84 g
1.16 g
Reflectivity R
R = 0.998
R = 0.952
Sail size D
≈ 0 at the optimal sail size
input at its 5th percentile
input at its 95th percentile
21
22
23
24
25
26
27
Travel time to α Cen [yr] (relativistic; centre = 23.6 yr at mean inputs)
b
Monte Carlo, N = 100,000
Classical: median 20.5 yr, 90% [18.1, 22.9]
Relativistic: median 23.6 yr, 90% [21.2, 26.0]
Probability density [1/yr]
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
16
18
20
22
24
26
28
30
Travel time to α Cen [yr] (Earth frame)